

1. Administrative & Study Identification

Full Study Title A Study to Determine the Efficacy and Safety of Deucravacitinib Compared With Placebo in Participants With Active Psoriatic Arthritis (PsA) Who Are Naïve to Biologic Disease Modifying Anti-rheumatic Drugs or Had Previously Received TNFα Inhibitor Treatment (Official: A Multi-center, Randomized, Double-blind, Placebo-controlled Phase 3 Study to Evaluate the Efficacy and Safety of Deucravacitinib in Participants with Active Psoriatic Arthritis (PsA) who are Naïve to Biologic Disease Modifying Anti-rheumatic Drugs or had Previously Received TNFα Inhibitor Treatment) **Short Title/Acronym** POETYK PsA-2 **Protocol Number** IM011-055 **NCT Number** NCT04908189 **Sponsor Name / Company** Bristol Myers Squibb **Investigational Product** Deucravacitinib (Sotyktu) / BMS-986165 **Phase of Development** PHASE3 **Trial Status** ACTIVE_NOT_RECRUITING **Medical Monitor** Edgar Charles, MD (VP and Senior Global Program Lead, Immunology) / Head of Global Clinical Trial Submission Unit

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and efficacy of deucravacitinib (6 mg once daily) versus placebo in achieving an ACR20 response at 16 weeks in participants with active psoriatic arthritis (PsA) who are naïve to biologic disease-modifying antirheumatic drugs (bDMARDs) or had previously received a TNFα inhibitor. The long term extension period will provide additional long-term safety and efficacy information. **Secondary Objectives:** To evaluate improvements in PsA disease activity at 16 weeks across clinical signs and symptoms, extra-articular manifestations, and patient-reported outcomes, including PASI 75 response, DAS28-CRP, HAQ-DI, Minimal Disease Activity (MDA), ACR50, ACR70, and improvements in the SF-36 physical component summary (PCS) score.

2.2 Background & Rationale To address the significant unmet needs of patients living with active psoriatic arthritis. Deucravacitinib is a first-in-class, oral, selective, allosteric tyrosine kinase 2 (TYK2) inhibitor that limits cytokine signaling involved in psoriasis and PsA pathogenesis. The POETYK PsA-2 study aims to confirm the efficacy and safety of this novel mechanism as an important oral treatment option for patients who have not achieved adequate responses with conventional therapies.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled and Active Comparator (Apremilast) **Blinding:** Double-blind **Randomization:** 3:3:1 (Deucravacitinib 6 mg once daily : Placebo : Apremilast 30 mg twice daily)

3.2 Planned Enrollment & Duration Targeted Enrollment: 729 participants **Number of Sites:** 144 locations **Estimated Study Start:** 15-Jul-2021 **Estimated Primary Completion:** 06-Feb-2024

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Diagnosed to have psoriatic arthritis (PsA) of at least 3 months duration at Screening. 2. Meets the Classification Criteria for Psoriatic Arthritis (CASPAR) criteria at Screening. 3. Active plaque psoriatic skin lesion(s) or documented medical history of plaque psoriasis (PsO) at Screening. 4. Active arthritis as shown by ≥ 3 swollen joints and ≥ 3 tender joints at Screening and Day 1. 5. Participant has high sensitivity C-reactive protein (hsCRP) ≥ 3 mg/L at Screening. 6. Must have completed the week 52 treatment for the optional open-label long-term extension period.

4.2 Exclusion Criteria 1. Nonplaque psoriasis at Screening or Day 1. 2. Other autoimmune conditions such as systemic lupus erythematosus, mixed connective tissue disease, multiple sclerosis, or vasculitis. 3. History of or current inflammatory joint disease other than PsA (e.g., gout, reactive arthritis, rheumatoid arthritis, ankylosing spondylitis, Lyme disease). 4. Active fibromyalgia. 5. Received an approved or investigational biologic therapy for the treatment of PsA or PsO. (*Note: Other protocol-defined inclusion/exclusion criteria apply*)

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Deucravacitinib 6 mg once daily; Placebo once daily; or Apremilast (Otezla; ATC Code: L04AA32) 30 mg twice daily. **Route of Administration:** Oral **Duration of Treatment:** 52-Week Treatment Period (consisting of a 16-week placebo-controlled period followed by a 36-week active treatment period where placebo patients are reallocated to deucravacitinib), followed by an optional open-label long-term extension.

5.2 Concomitant & Prohibited Therapy Permitted: Stable doses of non-steroidal anti-inflammatory drugs (NSAIDs) and conventional synthetic DMARDs as per standard of care. **Prohibited:** Biologic DMARDs (bDMARDs) during the study treatment period.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Up to 28 Days	Informed Consent, Eligibility, Joint Counts, CASPAR assessment
Baseline	Day 0/Day 1	Randomization, First Dose, Disease Activity Assessments
Interim 1	Week 16	Primary Efficacy Assessment (ACR20), Placebo group reallocated to deucravacitinib
End of Study	Week 52	Final 52-week assessments, Transition to long-term extension or enter follow-up
Follow-up	30 Days post-dose	Safety Follow-up Visit 30 days following the last dose

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Number of Participants Meeting American College of Rheumatology 20 (ACR20) response at Week 16. **Measurement Method:** Defined as a $\geq 20\%$ improvement over baseline in 68 tender and 66 swollen joint counts, and a $\geq 20\%$ improvement in 3 of the 5 remaining core data set measures: Participant Global Assessment of Disease Activity, Participant Global Assessment of Pain, Participant assessment of physical function, Physician Global Assessment of psoriatic arthritis, and Acute phase reactant value of high sensitivity C-reactive protein (hsCRP). Baseline is defined as the last measurement on or prior to date/time of first dose of study treatment.

7.2 Secondary & Exploratory Endpoints

- * Change From Baseline in Disease Activity Score 28 C-reactive Protein (DAS28-CRP) at Week 16.
- * Change From Baseline in Health Assessment Questionnaire - Disability Index (HAQ-DI) at Week 16.
- * Number of Participants Meeting Psoriasis Area and Severity Index 75 (PASI 75) ($\geq 75\%$ improvement from baseline) at Week 16.
- * Achievement of Minimal Disease Activity (MDA) at Week 16.
- * Improvement in physical function assessed by the SF-36 Physical Component Summary (PCS).
- * Resolution of enthesitis by Leeds Enthesitis Index (LEI).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection of clinical and laboratory AEs throughout the study period, including at-home monthly pregnancy testing between visits and evaluation of SARS-CoV-2-related AEs. **Serious Adverse Events (SAEs):** Monitored actively and evaluated for safety up to 30 days following the last dose of study treatment. **Data Monitoring Committee (DMC):** Supervised by the sponsor, Bristol Myers Squibb.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for efficacy outcomes. **Sample Size Justification:** 729 patients properly powered to detect superiority of deucravacitinib over placebo in achieving an ACR20 response at Week 16. **Handling of Missing Data:** Standard methodologies such as Non-Responder Imputation (NRI) for categorical binary endpoints (e.g., ACR20/50/70, PASI 75) and appropriate mixed models for continuous data.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine clinical site monitoring, centralized data review, and completion of standardized patient-reported outcome questionnaires (e.g., PHQ-8, eC-SSRS available for unscheduled safety visits). **Audit Trail:** Electronic Case Report Forms (eCRF), strict database locking mechanisms, and continuous medical review protocols.

11. Study Identifiers & Governance

Primary ID: IM011-055 **Posting ID:** 841936573973242978 **Registry Number:** NCT04908189 **Sponsor/Lead Organization:** Bristol Myers Squibb **Study Title:** POETYK PsA-2 **Trial Status:** ACTIVE_NOT_RECRUITING **Official Regulatory Title:** A Multi-center, Randomized, Double-blind, Placebo-controlled Phase 3 Study to Evaluate the Efficacy and Safety of Deucravacitinib in Participants with Active Psoriatic Arthritis (PsA) who are Naïve to Biologic Disease Modifying Anti-rheumatic Drugs or had Previously Received TNF α Inhibitor Treatment

12. Clinical Framework (Psoriatic Arthritis Specific)

Disease Areas / Conditions: Psoriatic Arthritis **Preferred UMLS Name:** Psoriatic Arthritis Drug Therapy **Diagnosis Threshold:** ≥ 3 swollen joints and ≥ 3 tender joints; hsCRP ≥ 3 mg/L **Medical Methodology:** Selective Tyrosine Kinase 2 (TYK2) inhibition **Optimization Target:** ACR20 response, DAS28-CRP, HAQ-DI, and PASI 75 achievement

13. Study Procedures & Methodology

Management Pathway: Oral systemic therapy for PsA management **Education Protocol (HCP):** Investigator meeting and protocol site training **Education Protocol (Patient):** Informed consent,

compliance training, and at-home pregnancy test kit instructions
Quality Audit Mechanism: Standard GCP clinical monitoring and patient questionnaire tracking

14. Observation & Follow-Up Schedule

Total Duration: 52-week treatment period + optional open-label long-term extension + 30 days safety follow-up **Onsite Visit Cadence:** Baseline, Week 16, Week 52, plus up to ~15 total hospital visits **Remote Monitoring Frequency:** Monthly at-home pregnancy testing **Checklist Review Interval:** Unscheduled visits as needed for safety/AE collection

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				